Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	21 October 2026
Team ID	
Project Name	Incident Lifecycle Management System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

Every incident moves through four stages (create, assign, resolve, close) on one ServiceNow cloud instance.

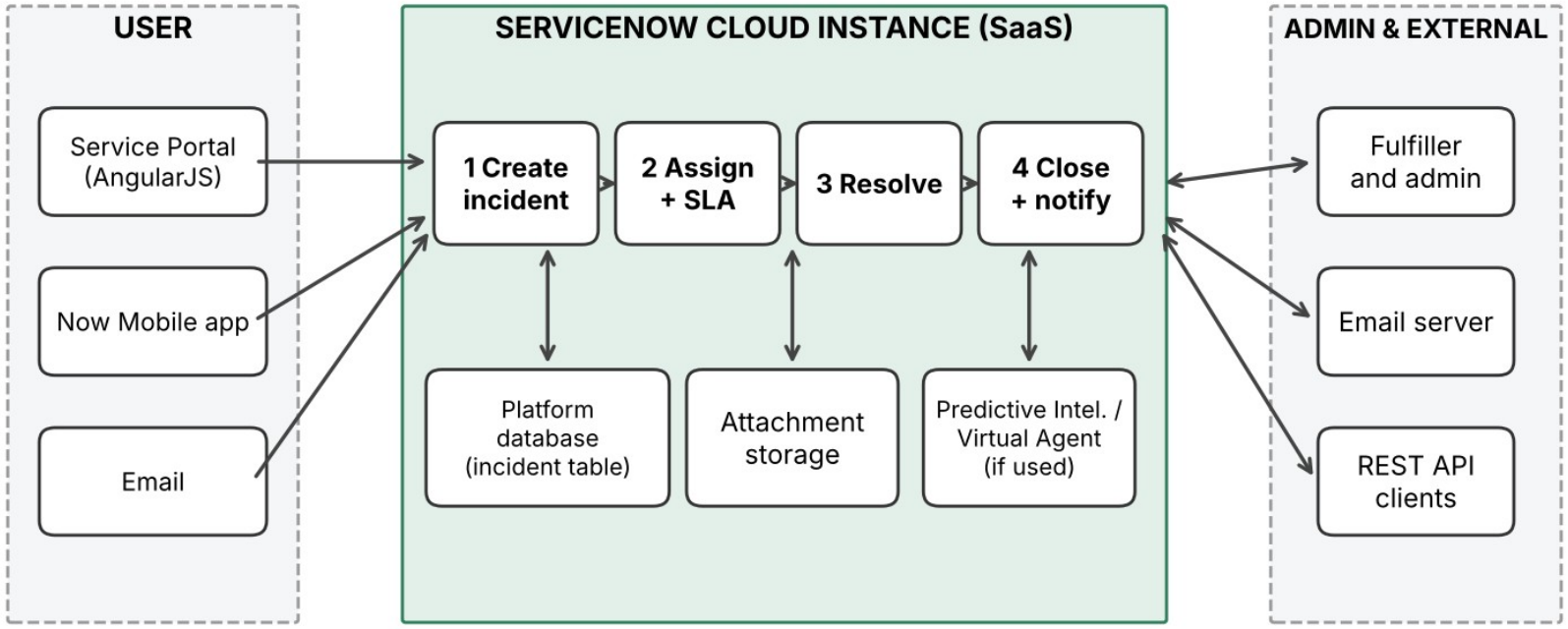


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web portal and mobile app where users raise and track incidents	ServiceNow Service Portal (HTML, CSS, JavaScript, AngularJS), Now Mobile app
2	Application Logic-1	Incident creation: validation, defaulting and form behaviour	Business Rules, Client Scripts, UI Policies
3	Application Logic-2	Assignment and escalation: routing to groups, SLA tracking	Assignment Rules, SLA Definitions, Flow Designer, Workflow
4	Application Logic-3	Resolution and closure: state changes, notifications, auto-close	Flow Designer, Notifications, Scheduled Jobs
5	Database	Incident data stored in relational tables (incident extends task), configurations	ServiceNow platform relational database
6	Cloud Database	Database service hosted on the cloud instance	ServiceNow cloud instance database (Personal Developer Instance)
7	File Storage	Attachments (screenshots, logs) on incident records	ServiceNow attachment storage (sys_attachment)
8	External API-1	Create/update incidents from email and send email notifications	Inbound Email Actions, SMTP/IMAP
9	External API-2	Integrate third-party systems with incident data	REST API / REST Message (JSON)
10	Machine Learning Model	Auto-categorise/route incidents and answer common queries (if used in project)	Predictive Intelligence, Virtual Agent
11	Infrastructure (Server / Cloud)	Application deployment on Cloud. Local Server Configuration: not required. Cloud Server Configuration: ServiceNow-hosted instance	ServiceNow cloud instance (SaaS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source front-end frameworks used in the portal	AngularJS, Bootstrap, jQuery
2	Security Implementations	Access control, authentication, encryption and auditing	ACLs, roles/groups, HTTPS/TLS, SSO (SAML) and MFA, audit/history logs
3	Scalable Architecture	Three-tier cloud architecture: presentation, application, database	ServiceNow multi-tier cloud architecture
4	Availability	Data-center replication, failover and managed backups	ServiceNow cloud infrastructure
5	Performance	Indexing, caching, scheduled jobs and asynchronous business rules	DB indexing, platform caching, scheduled jobs, async business rules